Confirmatory vs. Exploratory Analysis — Decision Guide

Fillable PDF. Click a blue box to type; tick the small squares. Save with **File › Save** and attach the saved PDF to your assignment. Works best in the free Adobe Acrobat Reader; macOS Preview and web browsers also work. If an answer does not fit its box, continue it on an extra page. Source: `Step_05_Preregistration_and_Open_Science/confirmatory_vs_exploratory_guide.md` (rebuild with `tools/make_fillable_pdfs.py`).

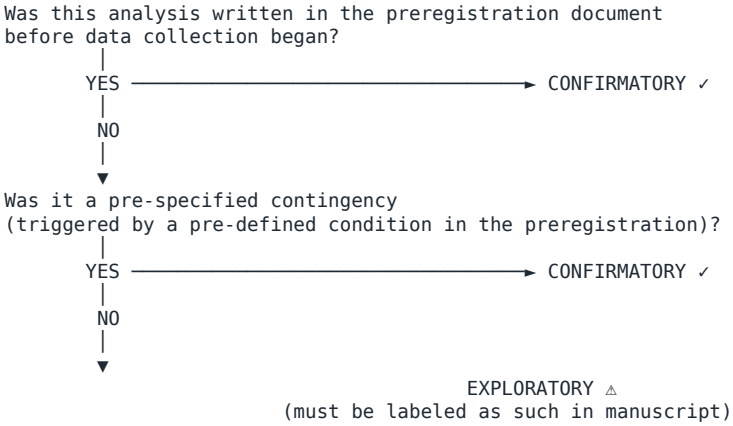
> Use this guide when writing the preregistration and again during analysis to classify every test or pattern you report. Mislabeling is the mechanism of HARKing: transparent labeling is the fix.

The core distinction

	Confirmatory	Exploratory
Hypothesis written	Before data collection	After seeing data (or not at all)
Analysis plan specified	Before data collection	During or after analysis
Inferential weight	High — justifies causal/directional claims	Low — hypothesis-generating only
Reported as	"We predicted… and found…"	"We observed… which may suggest…"
Requires replication?	Less urgently	Yes, before strong claims

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Decision flowchart



Frequently asked cases

Scenario	Classification	Rationale
Hypothesis H1 tested exactly as written in preregistration	**Confirmatory**	Pre-specified
Same H1 but you switched from OLS to a robust regression after seeing residuals	**Exploratory**	Analysis deviation — note as amendment
Subgroup analysis not mentioned in preregistration	**Exploratory**	Post-hoc
Subgroup analysis listed as contingency in Section 8.4 of preregistration	**Confirmatory**	Pre-specified contingency
You notice an unexpected correlation in the data and test it	**Exploratory**	Classic HARKing risk — must be labeled
Qualitative coding of open-ended items, planned in Section 8	**Confirmatory**	Pre-specified
Qualitative theme that emerged unexpectedly during coding	**Exploratory**	Post-hoc
Reliability check (Cronbach's α) run as an assumption check	**Confirmatory**	Pre-specified in Section 8.3

How to report in the manuscript

Confirmatory results section

> "We pre-registered the hypothesis that [X → Y]. Consistent with this prediction, [result, statistic, CI]." "We pre-registered the hypothesis that [X → Y]. Contrary to this prediction, [result, statistic, CI]."

Do not omit null confirmatory results. A null result on a pre-specified test is informative and must be reported.

Exploratory results section

> "In exploratory analyses, we observed [result]. This pattern was not pre-specified and should be interpreted with caution pending replication."

Common HARKing warning signs (self-audit)

- [ ] Did you run any analyses *before* finalizing which ones to call "confirmatory"?
- [ ] Did the hypothesis wording change after you saw the direction of results?
- [ ] Did you add or remove covariates after inspecting the outcome variable?
- [ ] Did you change the operationalization of a variable after seeing correlations?
- [ ] Did you drop a hypothesis because it was not significant, without reporting it?

If you checked any box above, those analyses **must** be labeled exploratory.

Amendment procedure

If you need to deviate from the preregistration *during* analysis (e.g., a measure has inadequate reliability):

1. **Stop analysis** before examining the outcome variable.
2. Write a timestamped amendment in Section 11 of `preregistration_template.md`.
3. Upload the amendment to OSF/AsPredicted.
4. Proceed. The amended analysis retains exploratory status unless it was a pre-specified contingency.